Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Manuel Mendes Dairy (Golden M)

Physical address of dairy:

29744 Cottonwood RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X250-X008-XXXX X063-X250-X030-XXXX

B. OPERATORS

Mendes, Manuel

Operator name: Mendes, Manuel Telephone no.: (209) 854-9108 (Landline) (209) 587-0708 (Cellular)

29744 Cottonwood RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Mendes, Manuel

Legal owner name: Mendes, Manuel Telephone no.: (209) 854-9108 (Landline) (209) 587-0708 (Cellular)

29744 Cottonwood RD	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	200	0	0	0
Number under roof	600	90	0	0	0	100
Maximum number	600	90	200	0	0	120
Average number	600	90	200	0	0	100
Avg live weight (lbs)	1,400	1,450	1,100	0		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,350.70 *tons per reporting period*

Total nitrogen from manure: 243,584.41 *lbs per reporting period*

After ammonia losses (30% loss applied): 170,509.09 *lbs per reporting period*

Total phosphorus from manure: 40,146.72 *lbs per reporting period*

Total potassium from manure: 115,693.94 *lbs per reporting period*

Total salt from manure: 303,205.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 22,416,000 *gallons*

Total nitrogen generated: 148,399.87 *lbs*

Total phosphorus generated: 12,480.05 *lbs*

Total potassium generated: 205,706.79 *lbs*

Total salt generated: 1,380,057.38 *lbs*

	22,416,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	22,416,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	Corral solids *Solid Manure*	760.00 *ton*	Dry-weight	71.0	15,100.00	4,630.00	37,700.00		62.40

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
08/03/2023	Solid commercial fertilizer *UN32*	1.98 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,265.93	0.00	0.00	0.00
Dry manure	6,656.08	2,040.90	16,618.16	275,059.20
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	7,922.01	2,040.90	16,618.16	275,059.20

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/09/2023	Corral solids	300.00 *ton*	Dry-weight	69.5		21,800.00	4,700.00	19,700.00		38.80

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	3,989.40	860.10	3,605.10	71,004.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	3,989.40	860.10	3,605.10	71,004.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy	101	101	2	process wastewater	X063-X250-X008-XXXX X063-X250-X009-XXXX
Heifer Ranch	128	128	2	process wastewater	X069-X290-X021-XXXX X069-X290-X022-XXXX X069-X290-X033-XXXX XX69-X290-X034-XXXX
McVicker	75	75	1	process wastewater	X069-X290-X020-XXXX
Pasture	15	15	0	none	X070-X050-X015-XXXX
Pometta	53	53	2	process wastewater	X069-X290-X031-XXXX X069-X290-X033-XXXX XX69-X290-X034-XXXX
Scotty's	3	3	0	none	X070-X020-X046-XXXX
Taglio	36	36	2	manure	X069-X280-X014-XXXX
Walnut	16	16	2	process wastewater	X063-X250-X030-XXXX
Totals for areas that were used for application	409	409	11		
Totals for areas that were not used for application	18	18	0		
Land application area totals	427	427	11		

B. CROPS AND HARVESTS

Dairy

Field name: Dairy

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 101 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	2,171.00 *ton*	As-is		74.2	3,260.00	650.00	5,610.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.50	140.15	27.94	241.17	1,308.79

Dairy

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 101 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	2,469.00 *ton*	As-is		72.0	3,640.00	530.00	3,250.00		2.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.45	177.96	25.91	158.90	383.31

Heifer Ranch

Field name: Heifer Ranch

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 128 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	2,529.00 *ton*	As-is		73.6	3,610.00	640.00	5,340.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.76	142.65	25.29	211.01	1,293.58

07/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 128 Plant date: 07/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	3,045.00 *ton*	As-is		72.8	3,730.00	510.00	3,080.00		2.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.79	177.47	24.26	146.54	310.59

McVicker

Field name: McVicker

11/17/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 75 Plant date: 11/17/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	581.00 *ton*	As-is		9.0	26,600.00	1,870.00	19,600.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.75	412.12	28.97	303.67	1,226.61

Pometta

Field name: Pometta

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 53 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,125.00 *ton*	As-is		74.0	3,350.00	640.00	5,430.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.23	142.22	27.17	230.52	1,468.02

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,136.00 *ton*	As-is		70.2	3,810.00	520.00	3,360.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	21.43	163.33	22.29	144.04	740.93

Taglio

Field name: Taglio

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	854.00 *ton*	As-is		75.0	3,020.00	670.00	5,850.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	23.72	143.28	31.79	277.55	1,482.64

07/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 07/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	872.00 *ton*	As-is		71.0	3,360.00	500.00	3,160.00		2.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.22	162.77	24.22	153.08	365.27

Walnut

Field name: Walnut

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	351.00 *ton*	As-is		73.8	3,760.00	730.00	6,310.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	21.94	164.97	32.03	276.85	1,517.37

Walnut

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	375.00 *ton*	As-is		71.0	3,830.00	480.00	3,080.00		2.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.44	179.53	22.50	144.38	394.22

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy - 11/02/2022: Oats, silage-soft dough

Field name: Dairy

Crop: Oats, silage-soft dough — Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.51	12.30	118.91	845.22	1,536,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	697.90	17,064,000.00 *gal*
Application event totals		83.51	12.30	118.91	1,543.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.32	4.61	44.59	316.96	576,000.00 *gal*
Application event totals		31.32	4.61	44.59	316.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.00	6.29	98.99	742.42	1,152,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	213.00	16,632,000.00 *gal*
Application event totals		77.00	6.29	98.99	955.42	

Dairy - 06/22/2023: Corn, silage

Field name: Dairy

Crop: Corn, silage — Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dairy - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	118.98	6.23	166.57	1,073.18	1,440,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	237.90	18,576,000.00 *gal*
Application event totals		118.98	6.23	166.57	1,311.08	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	219.92	17,172,000.00 *gal*
Application event totals		0.00	0.00	0.00	219.92	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.87	5.24	94.80	571.10	1,152,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	215.77	16,848,000.00 *gal*
Application event totals		65.87	5.24	94.80	786.86	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	221.30	17,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	221.30	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	213.00	16,632,000.00 *gal*
Application event totals		0.00	0.00	0.00	213.00	

Dairy - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.40	3.93	71.10	428.32	864,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	210.23	16,416,000.00 *gal*
Application event totals		49.40	3.93	71.10	638.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	208.85	16,308,000.00 *gal*
Application event totals		0.00	0.00	0.00	208.85	

Heifer Ranch - 11/19/2022: Oats, silage-soft dough

Field name: Heifer Ranch

Crop: Oats, silage-soft dough Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.84	14.56	140.75	1,000.40	2,304,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	683.13	21,168,000.00 *gal*
Application event totals		98.84	14.56	140.75	1,683.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.14	7.45	117.16	878.73	1,728,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	208.45	20,628,000.00 *gal*
Application event totals		91.14	7.45	117.16	1,087.18	

Heifer Ranch - 07/05/2023: Corn, silage

Heifer Ranch - 07/05/2023: Corn, silage

Field name: Heifer Ranch

Crop: Corn, silage

Plant date: 07/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.66	5.90	157.72	1,016.17	1,728,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	202.99	20,088,000.00 *gal*
Application event totals		112.66	5.90	157.72	1,219.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.97	5.16	93.51	563.29	1,440,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	197.54	19,548,000.00 *gal*
Application event totals		64.97	5.16	93.51	760.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	189.90	18,792,000.00 *gal*
Application event totals		0.00	0.00	0.00	189.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.62	18,468,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.97	4.13	74.80	450.63	1,152,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	181.17	17,928,000.00 *gal*
Application event totals		51.97	4.13	74.80	631.80	

Heifer Ranch - 07/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	176.80	17,496,000.00 *gal*
Application event totals		0.00	0.00	0.00	176.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	170.25	16,848,000.00 *gal*
Application event totals		0.00	0.00	0.00	170.25	

McVicker - 11/17/2020: Alfalfa, hay

Field name: McVicker

Crop: Alfalfa, hay

Plant date: 11/17/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	192.27	10.07	269.18	1,734.26	1,728,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	203.02	11,772,000.00 *gal*
Application event totals		192.27	10.07	269.18	1,937.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	191.85	11,124,000.00 *gal*
Application event totals		0.00	0.00	0.00	191.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	195.57	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

McVicker - 11/17/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.26	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	282.29	16,368,000.00 *gal*
Application event totals		0.00	0.00	0.00	282.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	178.81	10,368,000.00 *gal*
Application event totals		0.00	0.00	0.00	178.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	175.08	10,152,000.00 *gal*
Application event totals		0.00	0.00	0.00	175.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.46	7.18	144.84	1,081.83	1,152,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	171.36	9,936,000.00 *gal*
Application event totals		114.46	7.18	144.84	1,253.19	

Pometta - 11/10/2022: Oats, silage-soft dough

Field name: Pometta

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Pometta - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.51	13.18	127.47	906.02	864,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	216.13	8,856,000.00 *gal*
Application event totals		89.51	13.18	127.47	1,122.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.83	7.99	125.76	943.21	768,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	210.86	8,640,000.00 *gal*
Application event totals		97.83	7.99	125.76	1,154.07	

Pometta - 06/25/2023: Corn, silage

Field name: Pometta

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.81	5.54	148.13	954.39	672,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	229.31	9,396,000.00 *gal*
Application event totals		105.81	5.54	148.13	1,183.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	221.40	9,072,000.00 *gal*
Application event totals		0.00	0.00	0.00	221.40	

Pometta - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.76	4.99	90.33	544.16	576,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	213.50	8,748,000.00 *gal*
Application event totals		62.76	4.99	90.33	757.65	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	216.13	8,856,000.00 *gal*
Application event totals		0.00	0.00	0.00	216.13	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.30	4.16	75.28	453.46	480,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	200.32	8,208,000.00 *gal*
Application event totals		52.30	4.16	75.28	653.78	

09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	200.32	8,208,000.00 *gal*
Application event totals		0.00	0.00	0.00	200.32	

10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	197.68	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	197.68	

Taglio - 11/19/2022: Oats, silage-soft dough

Field name: Taglio

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Taglio - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	190.44	45.76	126.40	0.00	480.00 *ton*
Application event totals		190.44	45.76	126.40	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	731.15	6,372,000.00 *gal*
Application event totals		0.00	0.00	0.00	731.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	217.30	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	217.30	

Taglio - 07/05/2023: Corn, silage

Field name: Taglio

Crop: Corn, silage

Plant date: 07/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure - Import	Corral solids	184.89	56.69	461.62	7,640.53	760.00 *ton*
Application event totals		184.89	56.69	461.62	7,640.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	228.95	6,372,000.00 *gal*
Application event totals		0.00	0.00	0.00	228.95	

Taglio - 07/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	209.54	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	217.30	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	217.30	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	213.42	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	213.42	

09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	201.78	5,616,000.00 *gal*
Application event totals		0.00	0.00	0.00	201.78	

10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	186.26	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Walnut - 11/10/2022: Oats, silage-soft dough

Walnut - 11/10/2022: Oats, silage-soft dough

Field name: Walnut

Crop: Oats, silage-soft dough — Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.84	14.56	140.75	1,000.40	288,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	752.83	2,916,000.00 *gal*
Application event totals		98.84	14.56	140.75	1,753.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.27	8.27	130.18	976.37	240,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	218.27	2,700,000.00 *gal*
Application event totals		101.27	8.27	130.18	1,194.64	

Walnut - 06/22/2023: Corn, silage

Field name: Walnut

Crop: Corn, silage — Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.14	5.25	140.20	903.26	192,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	253.20	3,132,000.00 *gal*
Application event totals		100.14	5.25	140.20	1,156.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	193.37	2,392,000.00 *gal*
Application event totals		0.00	0.00	0.00	193.37	

Walnut - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.30	5.51	99.74	600.84	192,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	200.81	2,484,000.00 *gal*
Application event totals		69.30	5.51	99.74	801.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	200.81	2,484,000.00 *gal*
Application event totals		0.00	0.00	0.00	200.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.30	5.51	99.74	600.84	192,000.00 *gal*
CCID	Surface water	0.00	0.00	0.00	157.16	1,944,000.00 *gal*
Application event totals		69.30	5.51	99.74	758.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	183.35	2,268,000.00 *gal*
Application event totals		0.00	0.00	0.00	183.35	

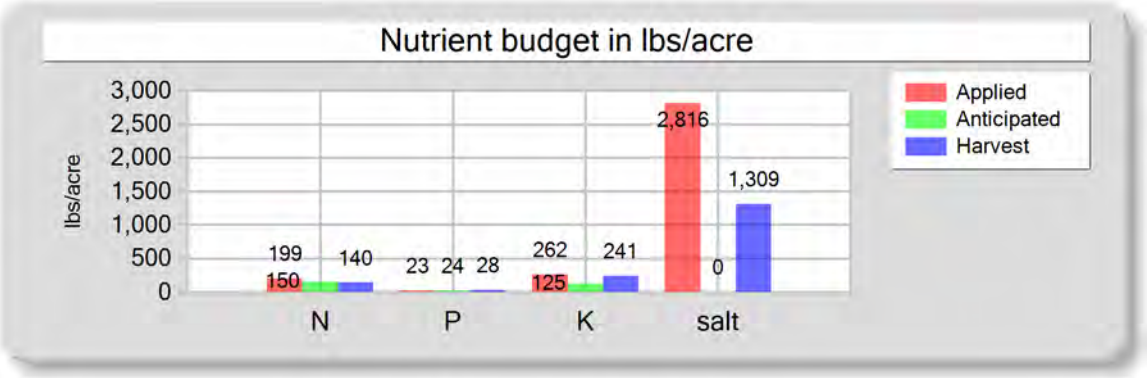
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Surface water	0.00	0.00	0.00	174.62	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	174.62	

B. NUTRIENT BUDGET

Dairy - 11/02/2022: Oats, silage-soft dough

Field name: Dairy | Crop: Oats, silage-soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.82	23.20	262.50	1,904.60
Fresh water	0.00	0.00	0.00	910.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.82	23.20	262.50	2,815.50
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	140.15	27.94	241.17	1,308.79
Nutrient balance	58.68	-4.74	21.32	1,506.71
Applied to removed ratio	1.42	0.83	1.09	2.15

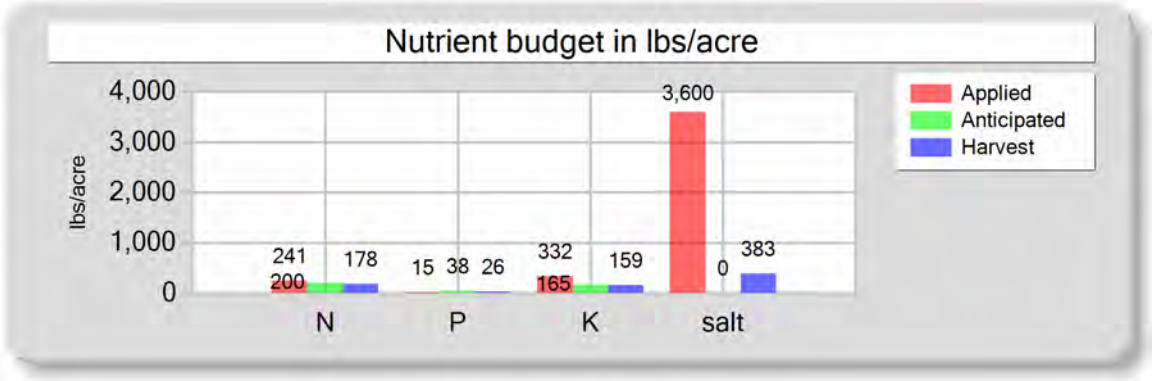
Fresh water applied	
33,696,000.00	*gallons*
1,240.91	*acre-inches*
12.29	*inches/acre*

Process wastewater applied	
3,264,000.00	*gallons*
120.20	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy - 06/22/2023: Corn, silage

Field name: Dairy | Crop: Corn, silage | Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	234.24	15.40	332.47	2,072.60
Fresh water	0.00	0.00	0.00	1,526.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.24	15.40	332.47	3,599.57
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	177.96	25.91	158.90	383.31
Nutrient balance	63.28	-10.52	173.58	3,216.26
Applied to removed ratio	1.36	0.59	2.09	9.39

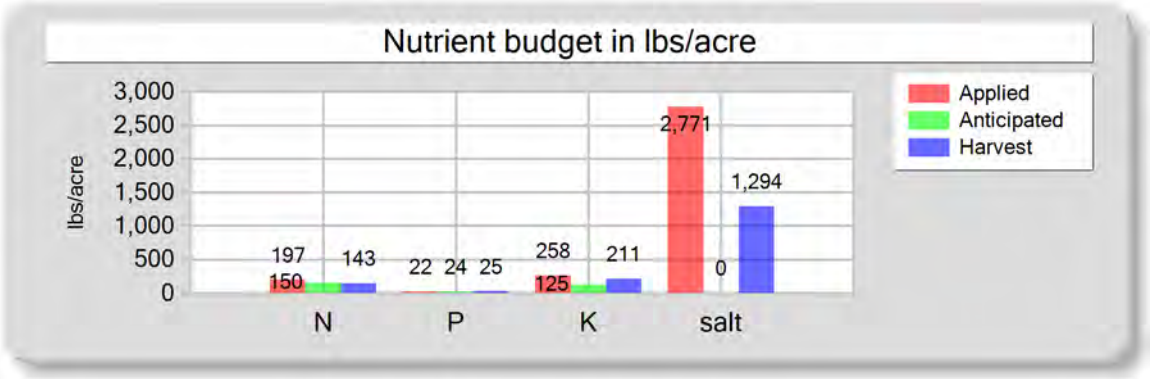
Fresh water applied	
119,232,000.00	*gallons*
4,390.91	*acre-inches*
43.47	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer Ranch - 11/19/2022: Oats, silage-soft dough

Field name: Heifer Ranch | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	189.98	22.00	257.91	1,879.13
Fresh water	0.00	0.00	0.00	891.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.98	22.00	257.91	2,770.71
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	142.65	25.29	211.01	1,293.58
Nutrient balance	54.33	-3.29	46.90	1,477.12
Applied to removed ratio	1.38	0.87	1.22	2.14

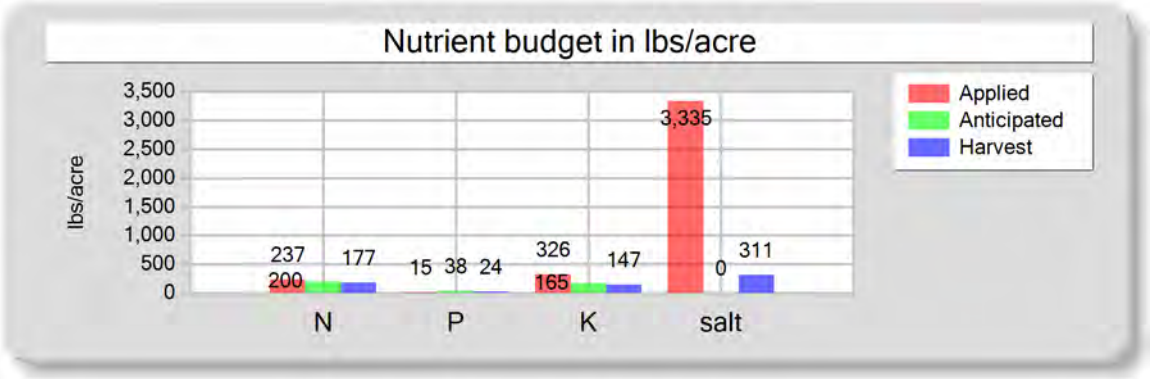
Fresh water applied	
41,796,000.00	*gallons*
1,539.20	*acre-inches*
12.03	*inches/acre*

Process wastewater applied	
4,032,000.00	*gallons*
148.48	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer Ranch - 07/05/2023: Corn, silage

Field name: Heifer Ranch | Crop: Corn, silage | Plant date: 07/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.60	15.20	326.03	2,030.09
Fresh water	0.00	0.00	0.00	1,305.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	236.60	15.20	326.03	3,335.37
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	177.47	24.26	146.54	310.59
Nutrient balance	59.13	-9.07	179.49	3,024.78
Applied to removed ratio	1.33	0.63	2.22	10.74

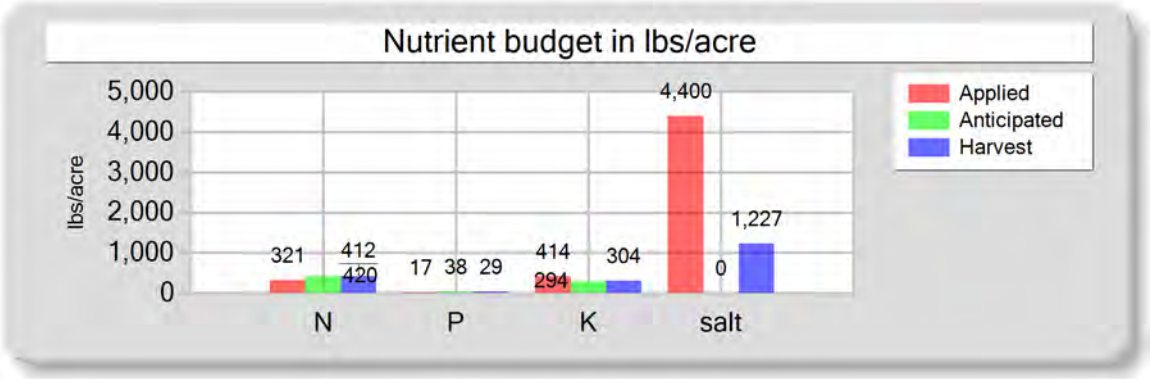
Fresh water applied	
129,168,000.00	*gallons*
4,756.82	*acre-inches*
37.16	*inches/acre*

Process wastewater applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

McVicker - 11/17/2020: Alfalfa, hay

Field name: McVicker Crop: Alfalfa, hay Plant date: 11/17/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	306.73	17.25	414.02	2,816.10
Fresh water	0.00	0.00	0.00	1,584.25
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	320.73	17.25	414.02	4,400.35
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	412.12	28.97	303.67	1,226.61
Nutrient balance	-91.39	-11.72	110.35	3,173.74
Applied to removed ratio	0.78	0.60	1.36	3.59

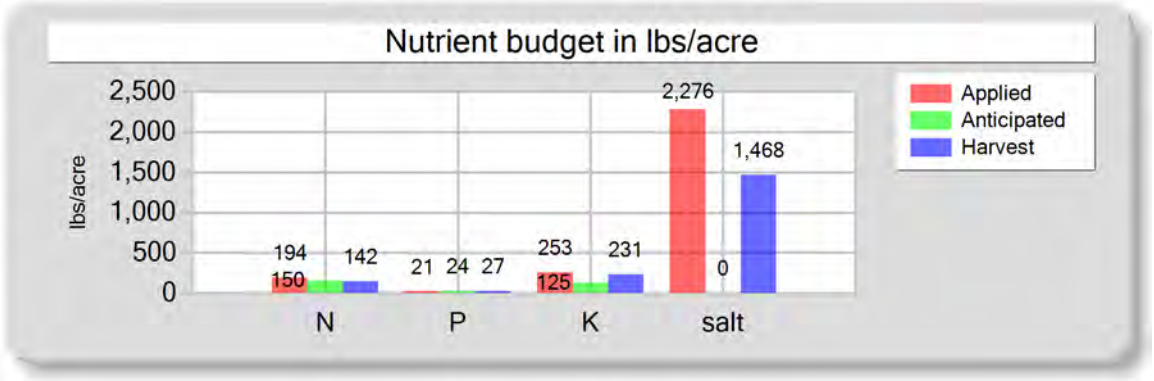
Fresh water applied	
91,860,000.00	*gallons*
3,382.89	*acre-inches*
45.11	*inches/acre*

Process wastewater applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pometta - 11/10/2022: Oats, silage-soft dough

Field name: Pometta | Crop: Oats, silage-soft dough | Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	187.34	21.18	253.23	1,849.23
Fresh water	0.00	0.00	0.00	426.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.34	21.18	253.23	2,276.22
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	142.22	27.17	230.52	1,468.02
Nutrient balance	52.12	-5.99	22.71	808.20
Applied to removed ratio	1.37	0.78	1.10	1.55

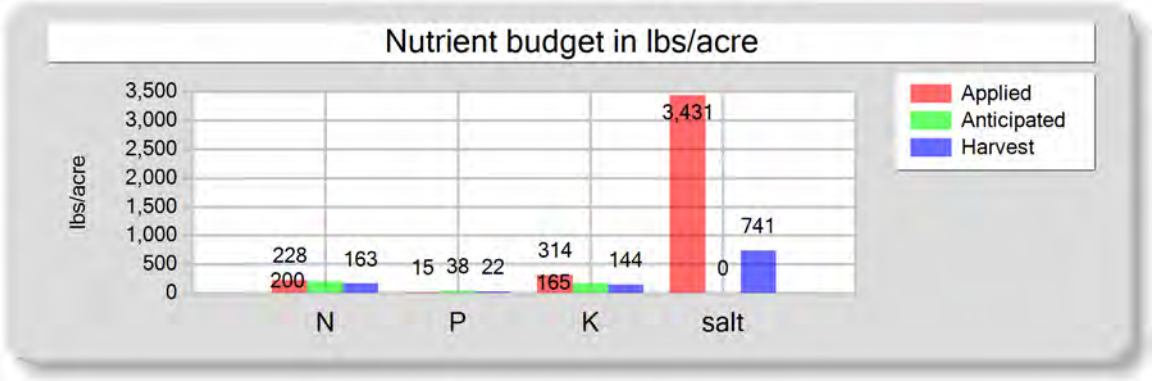
Fresh water applied	
17,496,000.00	*gallons*
644.32	*acre-inches*
12.16	*inches/acre*

Process wastewater applied	
1,632,000.00	*gallons*
60.10	*acre-inches*
1.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pometta - 06/25/2023: Corn, silage

Field name: Pometta | Crop: Corn, silage | Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.87	14.69	313.74	1,952.01
Fresh water	0.00	0.00	0.00	1,478.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.87	14.69	313.74	3,430.67
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	163.33	22.29	144.04	740.93
Nutrient balance	64.54	-7.60	169.70	2,689.74
Applied to removed ratio	1.40	0.66	2.18	4.63

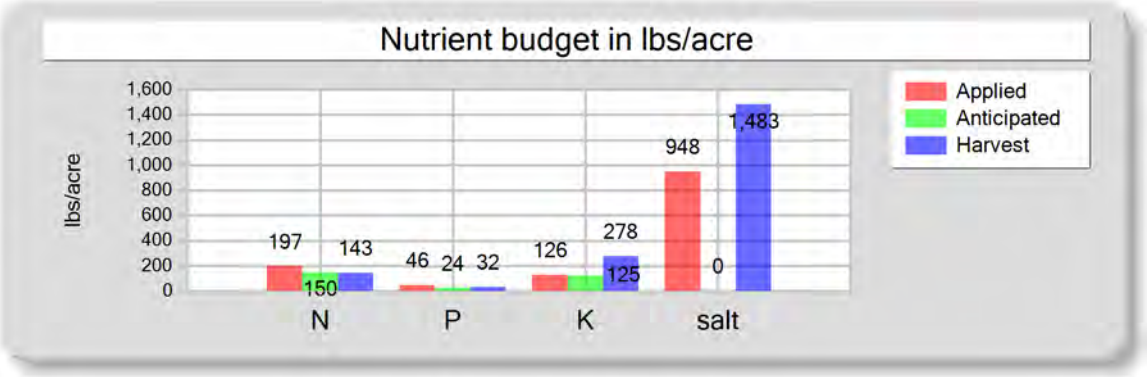
Fresh water applied	
60,588,000.00	*gallons*
2,231.25	*acre-inches*
42.10	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Taglio - 11/19/2022: Oats, silage-soft dough

Field name: Taglio | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	190.44	45.76	126.40	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	948.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.44	45.76	126.40	948.45
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	143.28	31.79	277.55	1,482.64
Nutrient balance	54.16	13.97	-151.15	-534.19
Applied to removed ratio	1.38	1.44	0.46	0.64

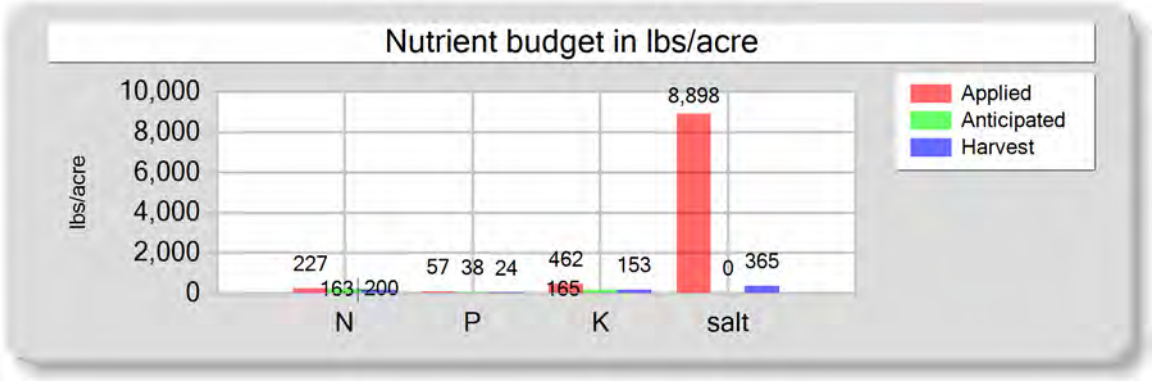
Fresh water applied	
12,420,000.00	*gallons*
457.39	*acre-inches*
12.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Taglio - 07/05/2023: Corn, silage

Field name: Taglio Crop: Corn, silage Plant date: 07/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	184.89	56.69	461.62	7,640.53
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,257.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.09	56.69	461.62	8,897.79
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	162.77	24.22	153.08	365.27
Nutrient balance	64.32	32.47	308.53	8,532.52
Applied to removed ratio	1.40	2.34	3.02	24.36

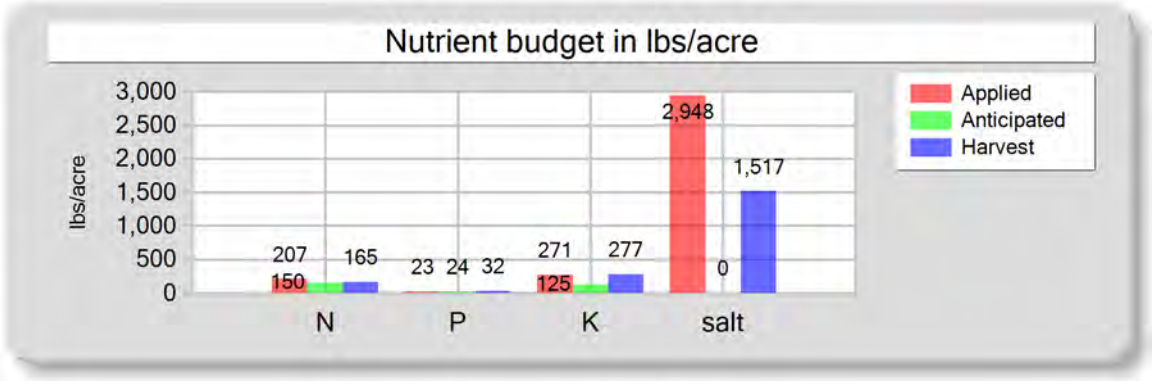
Fresh water applied	
34,992,000.00	*gallons*
1,288.64	*acre-inches*
35.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Walnut - 11/10/2022: Oats, silage-soft dough

Field name: Walnut | Crop: Oats, silage-soft dough | Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	200.10	22.83	270.93	1,976.76
Fresh water	0.00	0.00	0.00	971.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.10	22.83	270.93	2,947.87
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	164.97	32.03	276.85	1,517.37
Nutrient balance	42.13	-9.20	-5.92	1,430.50
Applied to removed ratio	1.26	0.71	0.98	1.94

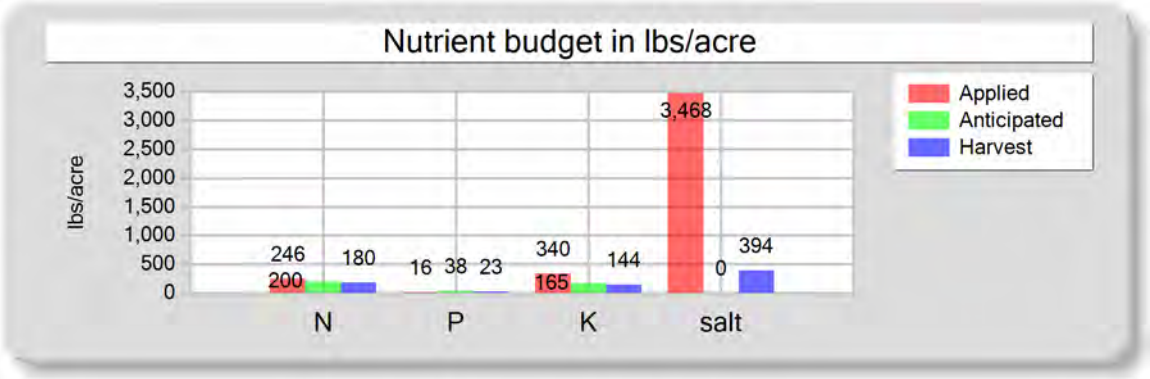
Fresh water applied	
5,616,000.00	*gallons*
206.82	*acre-inches*
12.93	*inches/acre*

Process wastewater applied	
528,000.00	*gallons*
19.44	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Walnut - 06/22/2023: Corn, silage

Field name: Walnut | Crop: Corn, silage | Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.73	16.26	339.67	2,104.94
Fresh water	0.00	0.00	0.00	1,363.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.73	16.26	339.67	3,468.27
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	179.53	22.50	144.38	394.22
Nutrient balance	66.20	-6.24	195.30	3,074.05
Applied to removed ratio	1.37	0.72	2.35	8.80

Fresh water applied	
16,864,000.00	*gallons*
621.04	*acre-inches*
38.82	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall Solid Manure

Sample and source description: Fall Solid Manure

Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	5,430.00	15,000.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Hfr Spring Solid Manure Import Sample

Sample and source description: Hfr Spring Solid Manure Import Sample

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	4,630.00	37,700.00	14,800.00	5,800.00	7,400.00	3,400.00	1,270.00		62.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Spring Solid Manure

Sample and source description: Spring Solid Manure

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	3,700.00	8,050.00	14,300.00	6,730.00	2,210.00	4,600.00	945.00		48.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Fall Solid Manure

Sample and source description: Fall Solid Manure

Sample date: 10/30/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,800.00	4,700.00	19,700.00	0.00	0.00	0.00	0.00	0.00		38.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	658.00	472.00	0.00	0.00	96.90	937.00								8,550.00	6,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	809.00	518.00	0.00	0.00	66.10	1,040.00								9,550.00	7,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,000.00	647.00	0.00	0.00	52.40	1,400.00								12,000.00	9,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 10/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	692.00	504.00	0.00	0.00	55.00	996.00								9,320.00	6,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	893.00	596.00	0.00	0.00	56.00	1,130.00	260.00	170.00	477.00	4,680.00	0.00	238.00	3.52	9,200.00	8,440
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

CCID

CCID

CCID

Sample description: CCID

Sample date: 08/17/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,940.00	495
DL	0.50	0.50	0.50								1.00	10

CCID

Sample description: CCID

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Dairy

Soil

Sample and source description: Soil

Sample date: 10/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			65.00				
DL			4.00				

Heifer Ranch

Heifer Ranch

Soil

Sample and source description: Soil

Sample date: 10/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			7.00				
DL			4.00				

McVicker

Soil

Sample and source description: Soil

Sample date: 10/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			15.00				
DL			4.00				

Pometta

Soil

Sample and source description: Soil

Sample date: 10/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			14.00				
DL			4.00				

Taglio

Taglio

Soil

Sample and source description: Soil

Sample date: 10/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			18.00				
DL			4.00				

Walnut

Soil

Sample and source description: Soil

Sample date: 10/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			71.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Dairy - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,260.00	650.00	5,610.00		11.80
DL	100.00	100.00	100.00		0.01

Dairy - 06/22/2023: Corn, silage

Dairy - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,640.00	530.00	3,250.00		2.80
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,480.00	520.00	3,220.00		2.30
DL	100.00	100.00	100.00		0.01

Heifer Ranch - 11/19/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage
Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,610.00	640.00	5,340.00		12.40
DL	100.00	100.00	100.00		0.01

Heifer Ranch - 07/05/2023: Corn, silage

Heifer Ranch - 07/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,730.00	510.00	3,080.00		2.40
DL	100.00	100.00	100.00		0.01

McVicker - 11/17/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 07/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,600.00	1,870.00	19,600.00		8.70
DL	100.00	100.00	100.00		0.01

Pometta - 11/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,350.00	640.00	5,430.00		13.30
DL	100.00	100.00	100.00		0.01

Pometta - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,810.00	520.00	3,360.00		5.80
DL	100.00	100.00	100.00		0.01

Taglio - 11/19/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,020.00	670.00	5,850.00		12.50
DL	100.00	100.00	100.00		0.01

Taglio - 07/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,360.00	500.00	3,160.00		2.60
DL	100.00	100.00	100.00		0.01

Walnut - 11/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,760.00	730.00	6,310.00		13.20
DL	100.00	100.00	100.00		0.01

Walnut - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,830.00	480.00	3,080.00		2.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

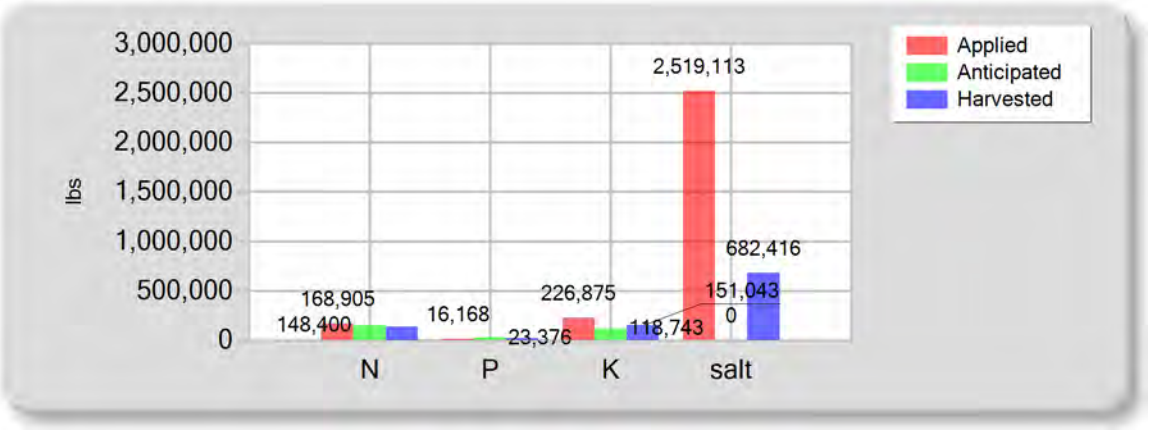
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

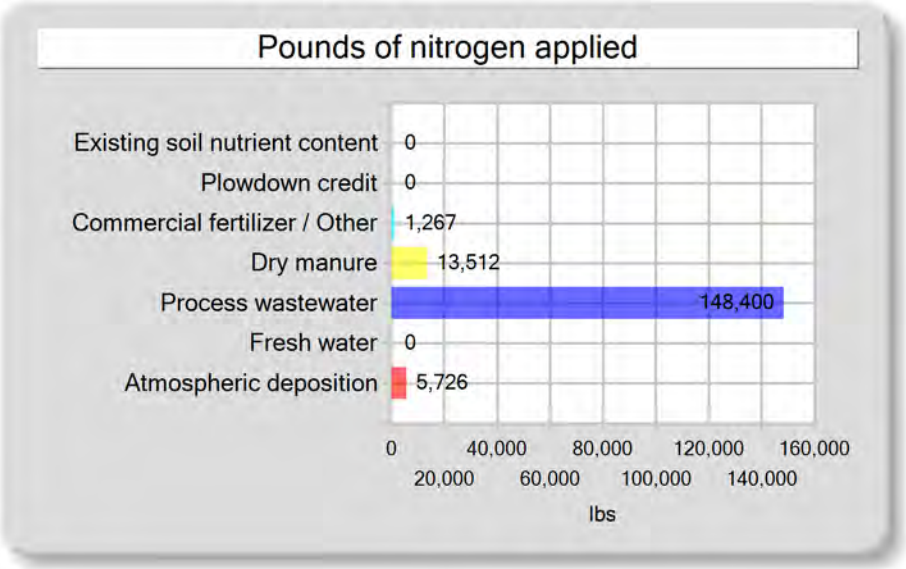
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

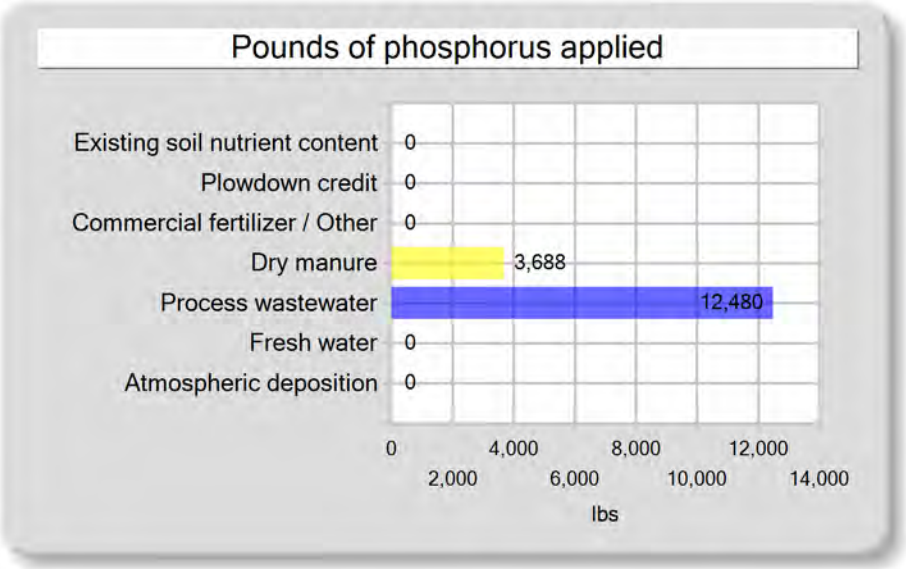
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,267.20	0.00	0.00	0.00
Dry manure	13,512.02	3,688.15	21,168.56	275,059.20
Process wastewater	148,399.87	12,480.05	205,706.79	1,380,057.38
Fresh water	0.00	0.00	0.00	863,996.57
Atmospheric deposition	5,726.00	0.00	0.00	0.00
Total nutrients applied	168,905.09	16,168.20	226,875.35	2,519,113.15
Anticipated crop nutrient removal	148,400.00	23,376.00	118,743.00	0.00
Actual crop nutrient removal	136,737.36	19,465.66	151,043.12	682,415.99
Nutrient balance	32,167.73	-3,297.46	75,832.23	1,836,697.16
Applied to removed ratio	1.24	0.83	1.50	3.69

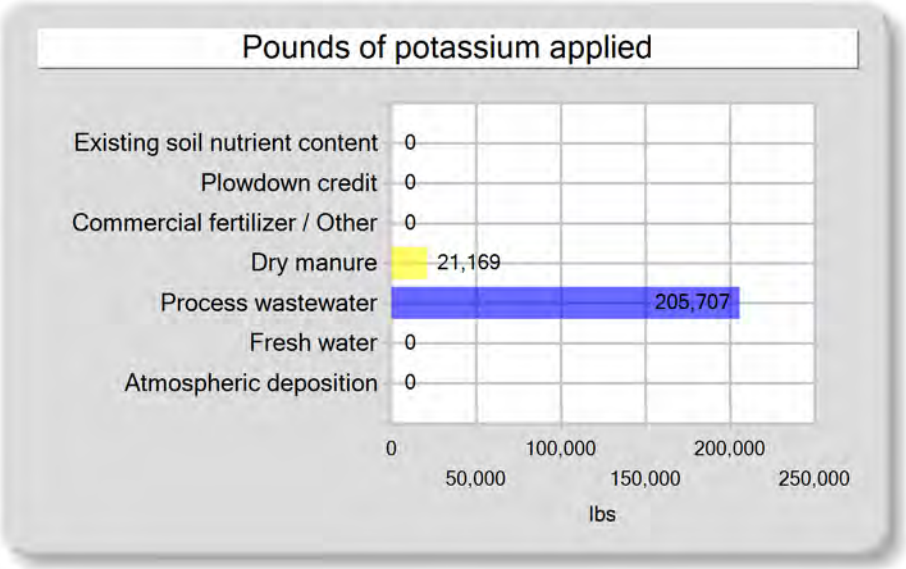
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

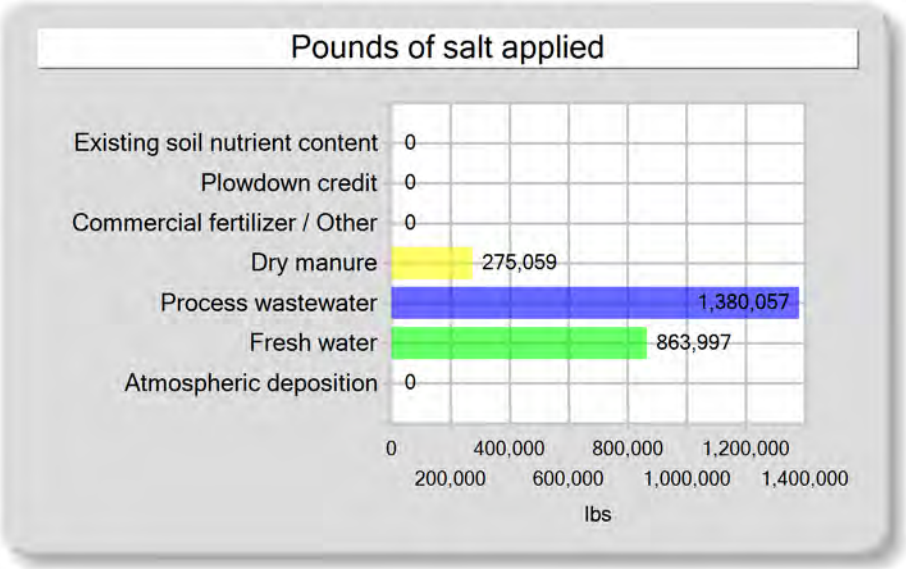


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Scott Mendes
Scott Mendes (May 10, 2024 11:18 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel Mendes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 10, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: MANUEL MENDES

Name of Dairy Facility: MANUEL MENDES DAIRY

Facility Address: 29744 COTTONWOOD RD. GUSTINE 95322
Number and Street City Zip Code

Contact Person Name and Phone Number: MANUEL MENDES 209-854-9108
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: J+F FERTILIZER

Address of Hauling Company /Person: 900 MEREDITH AVE. GUSTINE 95322
Number and Street City Zip Code

Contact Person: JOHN NUNES 209-854-6325
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

JOHN NUNES 900 MEREDITH AVE. GUSTINE 95322 209-854-6325
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

900 MEREDITH AVE. GUSTINE 95322
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 11/9/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 300 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 69.5% MOISTURE.
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

25 LOADS @ 12 TONS PER LOAD.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Scott Mendes
Scott Mendes (May 10, 2024 11:18 PDT) Date: May 10, 2024

Hauler's Signature: John Nunes Date: 4/4/24

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Manuel Mendes Dairy
29744 W. Cottonwood Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 7/25/23 1:15 PM
SUBMITTED BY: Sean
ANALYZED DATE: 7/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW House	T3206033A	Electrical Conductivity (EC)	EPA 120.1 and	2.06	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.14	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1320	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	97.3 (T3206002A)	97.8 (T3206002A)	0.552
Electrical Conductivity (EC)	99.8			0.351
Nitrate Nitrogen (NO3-N)	101	93.5 (T3206033A)	92.7 (T3206033A)	0.881

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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Compliance Analysis Report

Customer
Manuel Mendes Dairy
29744 W. Cottonwood Rd
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 7/25/23 1:15 PM
SUBMITTED BY: Sean
ANALYZED DATE: 7/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW Milk Barn	T3206033B	Electrical Conductivity (EC)	EPA 120.1 and	1.71	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.84	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1090	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	97.3 (T3206002A)	97.8 (T3206002A)	0.552
Electrical Conductivity (EC)	99.8			0.351
Nitrate Nitrogen (NO3-N)	101	93.5 (T3206033A)	92.7 (T3206033A)	0.881

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MANUEL MENDES
Mailing address:
(not facility address) GUSTINE CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	7/25/23	8:10	DW- HOUSE	DOMESTIC	20.2'		X				
2	"	8:15	DW - MILK BARN	"	20.4'		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	7-25-23 1:15			T3206033

Required to be submitted to any agency? (YES / NO)

2023 Manuel Mendes Annual Report

Final Audit Report 2024-05-10

Created:	2024-05-10
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA5MZrQKR2tZAQNvmZNVpMqBJI-DIziDgz

"2023 Manuel Mendes Annual Report" History

Document created by Sean Locke (slocke@fragservices.com)
2024-05-10 - 6:04:38 PM GMT

Document emailed to Scotty Mendes (scottymendes87@gmail.com) for signature
2024-05-10 - 6:06:02 PM GMT

Email viewed by Scotty Mendes (scottymendes87@gmail.com)
2024-05-10 - 6:06:13 PM GMT

Signer Scotty Mendes (scottymendes87@gmail.com) entered name at signing as Scott Mendes
2024-05-10 - 6:18:38 PM GMT

Document e-signed by Scott Mendes (scottymendes87@gmail.com)
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